10-5 Data Analysis [4]

Point!

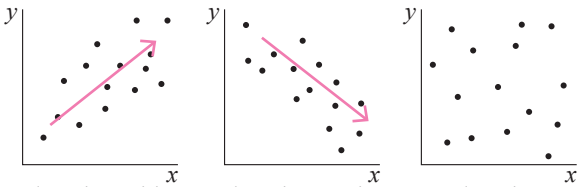
Scatter Plots, Correlations, and Cross Tabulation

(1) ([1]**Scatter plot**) : A graph that represents points on a plane using pairs of variables, x and y, as coordinates.

[1] When there is a tendency for both corresponding x and y to increase with each other, it is referred to as a ([2]**positive correlation**).

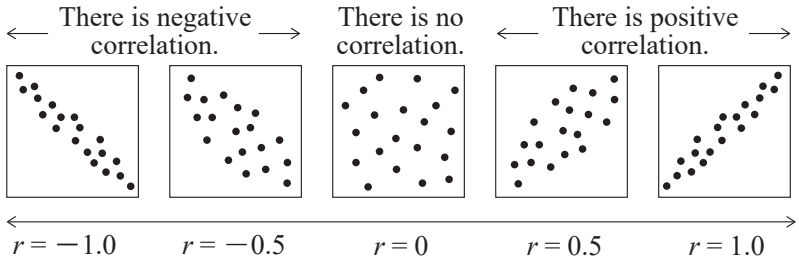
[2] When there is a tendency for one of the corresponding x and y to decrease as the other increases, it is referred to as a ([3]**negative correlation**).

[3] When there is neither a positive nor a negative correlation between the corresponding x and y, it is referred to as ([4]**no correlation**).



(2) ([5]**Correlation coefficient**) : An indicator to measure the strength of a correlation.

- The correlation coefficient r is represented by ([6] $-1 \leq r \leq 1$).
- The correlation coefficient r indicates a stronger ([7]**positive correlation**) as it approaches 1, and a stronger ([8]**negative correlation**) as it approaches -1.



- To determine the correlation coefficient using spreadsheet software, you can use the CORREL function.

(3) ([9]**Causal relationship**) : A relationship where one of two things is the cause and the other is the effect.
<Example> When the temperature rises, sales of shaved ice increase.

(4) ([10]**Pseudo-correlation**) : When two things appear to have a causal relationship, even though there is none.
<Example> There is no causal relationship between "As the temperature rises, shaved ice sales increase" and "As the temperature rises, the number of heatstroke cases increases."

(5) ([11]**Cross tabulation**) : An aggregation method for comparing data among two or more categories.

<Example>

	Agree	Oppose	Neither
Male	57	97	46
Female	73	75	52
Total	130	172	98

Warm Up

Answer the following questions.

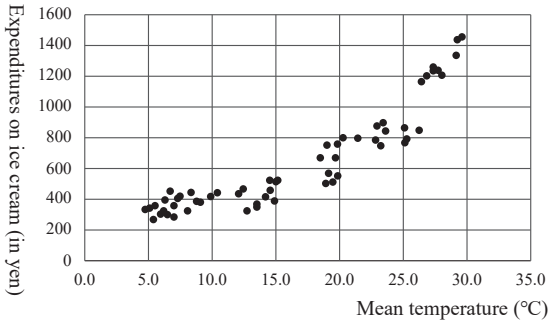
(1) There is data consisting of the following two variables, x and y.

x	7	4	6	2	9	3	8	1	6	3
y	3	6	5	8	3	4	2	9	2	5

[1] Create a scatter plot.

[2] Describe the tendency observed between x and y.

(2) The relationship between mean temperature and ice cream expenditures was investigated and summarized in the scatter plot on the right.



[1] Choose the description representing the tendency of the two from the options **A** to **D** below, and answer using the letter.

A There is a weak negative correlation.

B There is a strong negative correlation.

C There is a weak positive correlation.

D There is a strong positive correlation.

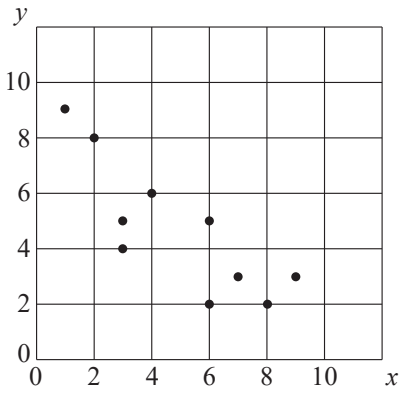
[2] Choose the number that best fits into the blank in the following sentence from the options **A** to **D** below, and answer using the letter.

The correlation coefficient between the mean temperature and ice cream expenditures was ( ).

A -0.9 **B** -0.4 **C** 0.4 **D** 0.9

Explanation

(1) [1] The scatter plot is shown below.



[2] There is a negative correlation according to the scatter plot.

(2) [1] Based on the scatter plot, it can be interpreted as having a “strong positive correlation” because it forms a linear upward graph. Therefore, **D**.

[2] Based on [1], since it is known that there is a strong positive correlation between mean temperature and ice cream expenditure, it can be inferred that the highest value of 0.9 is most appropriate. Therefore, **D**.

Try

1 There is data consisting of the following two variables, x and y.

x	7	4	6	2	8	3	8	1	6	3
y	3	6	5	8	3	4	2	8	2	5

(1) Create a scatter plot.

(2) Describe the tendency observed between x and y.

2 To examine the relationship between study time, commuting time, and test scores, a survey was taken by 40 students in a class regarding their daily study time on weekdays. Below are scatter plots representing the relationship between each student's study time and test scores (Figure 1), and the relationship between commuting time and test scores (Figure 2). Answer the following questions.

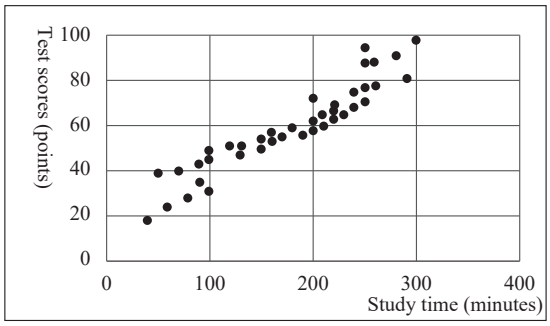


Figure 1: Scatter plot of study time and test scores

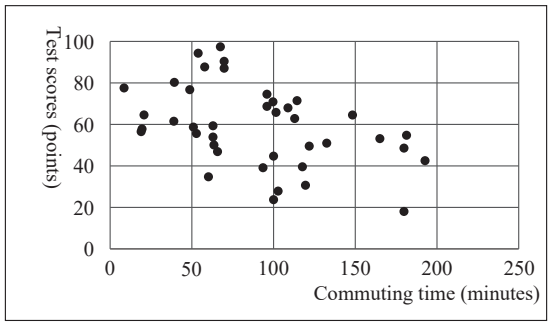


Figure 2: Scatter plot of commuting time and test scores

(1) Compare the two scatter plots, and choose the trend observed in each from the options **A** to **D** below, and answer using the letter.

A Figure 1 shows a strong positive correlation, while Figure 2 shows a strong negative correlation.

B Figure 1 shows a strong positive correlation, while Figure 2 shows a weak negative correlation.

C Figure 1 shows a weak positive correlation, while Figure 2 exhibits a strong negative correlation.

D Figure 1 shows a weak positive correlation, while Figure 2 shows a weak negative correlation.

(2) Choose the number that best fits into the blanks [1] and [2] in the following sentence from the options **A** to **D** below, and answer using the letters.

> When the correlation coefficients for study time, commuting time, and test scores are determined, the correlation coefficient between study time and test scores is ([1]), and the correlation coefficient between commuting time and test scores is ([2]).

A -0.9 **B** -0.4 **C** 0.4 **D** 0.9

3 The table on the right shows the results of a survey taken by 50 students in a class regarding whether they like Japanese, math, and English. Fill in the appropriate values for blanks [1] to [7] in the table.

	Like	Dislike	Neither	Total
Japanese	2	([1])	([2])	([3])
Math	6	([4])	5	13
English	([5])	([6])	1	18
Total	15	21	([7])	50

Exercise

1 There is data consisting of the following two variables, x and y.

x	6	4	7	5	8	6	2	9	8	5
y	7	10	2	4	3	4	4	5	4	7

(1) Create a scatter plot.

(2) Describe the tendency observed between x and y.

2 The rent of rental properties around a station was investigated to examine the relationship with the size of the room and the distance to the nearest station. The following are scatter plots representing the relationship between the size of each room and the rent (Figure 1), and the relationship between the distance to the nearest station from each room and the rent (Figure 2). Answer the following questions.

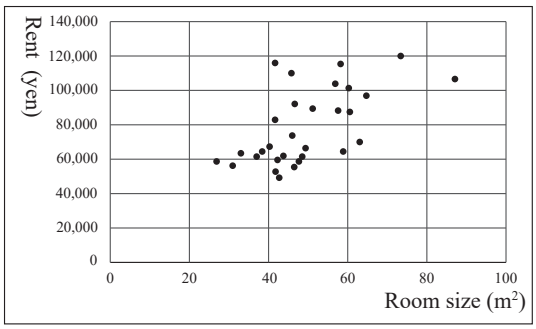


Figure 1: Scatter plot of room size and rent

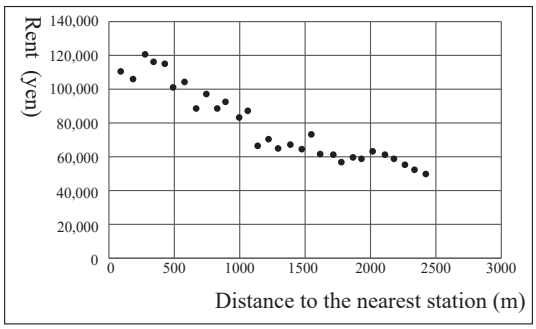


Figure 2: Scatter plot of rent and distance to nearest station

(1) Compare the two scatter plots and provide the tendencies that can be observed in each. Choose one from the options **A** to **D** below, and answer using the letter.

A Figure 1 shows a strong positive correlation, while Figure 2 shows a strong negative correlation.

B Figure 1 shows a weak positive correlation, while Figure 2 exhibits a strong negative correlation.

C Figure 1 shows a strong positive correlation, while Figure 2 shows a weak negative correlation.

D Figure 1 shows a weak positive correlation, while Figure 2 shows a weak negative correlation.

(2) Choose the number that best fits into the blanks [1] and [2] in the following sentence from the options **A** to **D** below, and answer using the letters.

> When the correlation coefficients for room size and rent, and distance to the nearest station and rent are determined, the correlation coefficient between room size and rent is ([1]), and the correlation coefficient between the distance to the nearest station and rent is ([2]).

A -0.9 **B** -0.6 **C** 0.6 **D** 0.9

3 The table on the right shows the results of a survey taken by 200 school students regarding their main means of commuting to and from school, categorized by grade. Fill in the appropriate values for blanks [1] to [7] in the table.

	On foot	Bicycle	Other	Total
Grade 1	29	33	([1])	([2])
Grade 2	([3])	([4])	([5])	67
Grade 3	([6])	32	9	66
Total	([7])	94	24	200